Nanyang Technological University
School of Electrical & Electronic Engineering
IE2106 Engineering Mathematics I

Tutorial 10 Laplace Transform Basics

1. Compute the Laplace Transforms *based on the definition of the Laplace Transform.*

   (a) $e^{-2t} \times u(t)$

discuss in class

   (b) $\sin 3t \times u(t)$

   (c) $t \times u(t)$

   (d) $(t-2) \times u(t-2)$

2. Sketch the function given by

   (a) $g(t) = 1$ for $0 \leq t \leq 2$
   $= 2$ for $2 \leq t \leq 3$
   $= 0$ elsewhere

   Write $g(t)$ in terms of (delayed) step functions.

   (b) What is the Laplace Transform of $g(t)$ in part (a)?

discuss in class

3. With the aid of basic properties and basic Laplace Transforms, find the Laplace Transforms of

   (a) $\sin(t-2)u(t-2)$

   (b) $e^{-(t-2)} \sin(t-2)u(t-2)$

   (c) $e^{-t} \sin(t-2)u(t-2)$

   (d) $\cos^2(t)u(t)$

4. Find the inverse Laplace Transform of

(a) $\dfrac{3s+2}{s(s+4)}$

discuss in class

(b) $\dfrac{2}{s(s+1)^2}$

(c) $\dfrac{2s+3}{s^2-2s+5}$

(d) $\dfrac{se^{-2s}}{s^2+16}$

Answers

1. (a) $\dfrac{1}{s+2}$ (b) $\dfrac{3}{s^2+9}$ (c) $\dfrac{1}{s^2}$ (d) $e^{-2s}\dfrac{1}{s^2}$

2. (a) $g(t) = u(t) + u(t-2) - 2u(t-3)$ (b) $\dfrac{1}{s}\left(1 + e^{-2s} - 2e^{-3s}\right)$

3. (a) $\dfrac{e^{-2s}}{s^2+a}$

(b) $\dfrac{e^{-2s}}{s^2+2s+2}$

(c) $\dfrac{e^{-2s-2}}{s^2+2s+2}$

(d) $\dfrac{1}{2s} + \dfrac{s}{2(s^2+4)}$

4. (a) $0.5u(t) + 2.5e^{-4t}u(t)$

(b) $(2 - 2te^{-t} - 2e^{-t})u(t)$

(c) $(2\cos 2t + 2.5\sin 2t)e^{t}u(t)$

(d) $\cos[4(t-2)]u(t-2)$